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A GROUP OF B7H3 MONOCLONAL ANTIBODIES AND MEDICAL USE THEREOF

Abstract

Provided are a group of anti-B7H3 monoclonal antibodies and humanized antibodies thereof. The antibodies obtained by means of hybridoma technology have B7H3 affinity, cell endocytosis effect, and antibody-dependent cytotoxicity, or can effectively block an immunosuppressive effect caused by B7H3, and can be used for preparing related drugs for inhibiting cancer cells, regulating the effect and level of B7H3, and enhancing the immunity of an organism.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Chinese Patent Application No. 202110639996.0, filed with the China National Intellectual Property Administration on Jun. 9, 2021 and entitled “A GROUP OF B7H3 MONOCLONAL ANTIBODIES AND MEDICAL USE THEREOF”, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure belongs to the field of anti-tumor antibody treatment and molecular immunology treatment, and relates to anti-B7H3 antibodies and use thereof. Specifically, the present disclosure relates to various anti-B7H3 monoclonal antibodies.

BACKGROUND

[0003] B7H3, also known as CD276, is a type I transmembrane protein (Chapoval A. I., et al., (2001), *Nat. Immunol.* 2:269). In mice, B7H3 is encoded by chromosome 9, while in humans, it is encoded by chromosome 15. B7H3 shares 20%-27% amino acid homology with other B7 family ligands (Sun M., et al., (2002), *J. Immunol.* 168:6294; Loos M., (2010), *Clin. Dev. Immunol.* 2010:683875). In terms of protein composition and structure, murine B7H3 has only two IgB7H3 types (consisting of a pair of extracellular IgV and IgC), and human B7H3 has four IgB7H3 types, which are almost identical tandem repeats of IgV-IgC (Sun M., et al., (2002), *J. Immunol.* 168:6294; Steinberger P., et al., (2004), *J. Immunol.* 172:2352). Although B7H3 mRNA is present in some normal tissues such as liver, small intestine, pancreas, testis, heart and colon, B7H3 protein is rarer in normal tissues (Greenwald R. J., et al., (2005), *Annu. Rev. Immunol.* 23:515-48). This difference between B7H3 mRNA and protein may reflect existence of close post-transcriptional regulatory mechanisms (Hofmeyer K. A., et al., (2008), *Proc. Natl. Acad. Sci.* 105:10277; Calabro L., et al., (2011), *J. Cell Physiol.* 226:2595). Generally, B7H3 is only present in some non-immune fibroblasts, endothelial cells, and osteoblasts, as well as some immune cells such as B cells, T cells, monocytes, dendritic cells, or NK cells. Expression of B7H3 can be induced by CSF or lipopolysaccharide stimulation in granulocyte-macrophage cells (Suh W. K., et al., (2003), *Nat. Immunol.* 4:899; Chapoval A. I., et al., (2001), *Nat. Immunol.* 2:269; Greenwald R. J. et al., (2005), *Annu. Rev. Immunol.* 23:515).

[0004] B7H3 belongs to important immune checkpoint proteins. Preliminary studies have shown that B7H3 has a stimulation effect on T cells. It can promote proliferation of CD4⁺ and CD8⁺ T cells to some extent, enhance the cytotoxicity of T lymphocytes, and stimulate production of interferon- γ (IFN- γ) (Chapoval A. I., et al., (2001), *Nat. Immunol.* 2:269). And deficiency of B7H3 can alleviate chronic rejection of cardiac allografts (Wang L. et al., (2005), *Eur. J. Immunol.* 35:428). However, more studies have shown that B7H3 also has significant immunosuppressive effects. For example, B7H3 can significantly inhibit activation effects of CD3 antibodies or allogeneic DC cells on T cells, and B7H3-blocking antibodies can effectively reverse the inhibitory effect (Prasad D. V. R., et al., (2004), *J. Immunol.* 173:2500). Structural information of B7H3 protein reveals that FG loop in B7H3 IgV domain may have a critical effect on inhibition of T cells (Vigdorovich V., et al., (2013), *Structure* 21:707). In addition to T cells, B7H3 may also have an inhibitory effect on natural killer cells (NK) (Castriconi R., et al., (2004), *Proc. Natl. Acad. Sci.* 101:12640). A possible mechanism is that regulatory T cells (Tregs) induce the expression of B7H3 in dendritic cells (DC), thereby inhibiting a stimulatory effect of DC cells on T lymphocytes

(Mahnke K., et al., (2007), *Eur. J. Immunol.* 37:2117). Mouse models lacking B7H3 exhibit more severe airway inflammations and earlier occurrence of some autoimmune diseases such as experimental autoimmune strain encephalomyelitis (Suh W. K., et al., (2003), *Nat. Immunol.* 4:899).

[0005] B7H3 is highly expressed in many malignant tumors such as melanoma (Wang J., et al., (2013), *J. Invest. Dermatol.* 133:2050), leukemia (Hu Y., et al., (2015), *Hematology* 20:187; Sun J., et al., (2014), *OncoTargets Ther.* 7:1979), prostate cancer (Zang X., et al., (2007), *Proc. Natl. Acad. Sci.* 104:19458), ovarian cancer (Zang X., et al., (2010), *Mod. Pathol.* 23:1104) and pancreatic cancer (Chen Y., et al., (2014), *Onco. Targets Ther.* 7:1465-72). Although B7H3 mRNA exists in both normal and tumor tissues, B7H3 protein levels are higher in only tumor cells. This difference may be related to a regulatory mechanism of miRNA-29. For example, studies have shown that there is an inverse relationship between miRNA-29 and B7H3 protein level in normal or tumor tissues and cancer cell lines. (Xu H., et al., (2009), *Cancer Res.* 69:6275). In tumor patients, abnormal expression of B7H3 is closely associated with poor prognosis, increased tumor grade and metastasis, treatment resistance, and decreased overall survival rate of tumor patients (Picarda E., et al., (2016), *Clin. Cancer Res.* 22:3425).

[0006] The inhibitory effect of B7H3 on T cells, NK cells and DC cells significantly promotes immune evasion of tumor cells. In addition, B7H3 plays an important role in proliferation, migration, invasion, angiogenesis, epithelial to mesenchymal transition (EMT), cancer stemness and Warburg effect of tumor cells and tumor cell drug resistance. An increase in B7H3 can promote expression of Bcl-2 and Bcl-xl, to activate a JAK2-STAT3 signaling pathway, thereby enhancing anti-apoptotic activity of tumor cells (Zhang T., et al., (2015), *World J. Gastroenterol.* 21:1804). In addition, activation of AKT, ERK and JAK2/STAT3 pathways can induce expression of tumor cell metastasis-related protein factors, including MMP2, MMP9, CXCR4, and the like, thereby enhancing migration and aggressiveness of tumor cells (Tekle C., et al., (2013), *Int. J Cancer* 130:2282; Li Y., et al., (2017), *Oncotarget* 8:71725; Wang L., et al., (2013), *PLOS One* 8: e70689; Liu F., et al., (2015), *Mol. Med. Rep.* 12:5455). Multiple studies have shown that B7H3 in tumor endothelial cells can also induce activation of NF- κ B through a TLR4-dependent mechanism, which significantly increases expression levels of VEGF and IL-8 (Tekle C., et al., (2012), *Int. J. Cancer* 130:2282), thereby promoting aggressiveness and angiogenesis of tumors (Ferrara N., et al., (2002), *Nat. Rev. Cancer* 2:795). In addition, B7H3 can also regulate activity of tumor cells EMT and cancer stem cells by reducing expression of E-cadherin and increasing expression of N-cadherin, vimentin, CD133 and CD44 (Jiang B., et al., (2016), *Oncotarget* 7:731755). Because B7H3 can activate STAT3 signaling to increase expression of hexokinase 2, B7H3 also has an effect of promoting aerobic glycolysis in tumor cells (Shi T., et al., (2019), *Cell Death Dis.* 10:308). In addition, B7H3 inhibits activity of transcription factor NRF2, which increases levels of reactive oxygen species (ROS) and HIF1 α , thereby further enhancing the aerobic glycolysis effect of tumor cells and promoting growth of tumor cells (Lim S., et al., (2016), *Cancer Res.* 76:2231). An increasing number of studies have shown that inhibiting or lowering expression of B7H3 can increase response of tumor cells to some drugs, for example, drugs for inhibiting DNA replication (including alkylation drugs) and drugs for inhibiting PI3K/Akt/mTOR and Ras/Raf/MEK signaling proteins (Flem-Karlsen K., et al., (2017), *Pigment Cell Melanoma Res.* 30:467; Kasten B. B., et al., (2017), *Nucl. Med. Biol.* 47:23; Liu H., et al., (2011), *Mol. Cancer Ther.* 10:960).

[0007] Transplanting tumor cells into mice lacking B7H3 or treating tumor mice with B7H3 antibodies significantly inhibits tumor growth (Cai D., et al., (2020), *Cell. Mol. Immunol.* 17:227; Lee Y. H., et al., (2017), *Cell Res.* 27:1034), indicating that blocking B7H3 signaling can be used for tumor treatment. In addition, using B7H3 antibodies in combination with anti-PD-1 antibodies can cause dual blockade of B7H3 and PD-1, thereby exerting a synergistic anti-tumor effect (Lee Y. H., et al., (2017), *Cell Res.* 27:1034). Due to a significant difference between B7H3 expression levels of normal tissues and tumor tissues, tumor cells can be effectively killed through the ADCC

effect of B7H3 antibodies or toxin conjugation without causing excessive impact on normal tissues (Koenig S., et al., (2014), *Medicographia* 36:285). In summary, based on current research, B7H3 is a promising target for cancer treatment.

SUMMARY

[0008] In the present disclosure, a group of anti-B7H3 monoclonal antibodies were obtained through hybridoma technology. These antibodies have high B7H3 affinity, significant cellular internalization (allowing selective tumor killing or inhibition through toxin conjugation), significant antibody-dependent cytotoxicity (ADCC), or effective blocking of the immunosuppressive action of B7H3. In the present disclosure, six candidate antibodies were successfully humanized. These antibodies have extensive potential in developing drugs to inhibit cancer cells, regulate the action and levels of B7H3, and enhance immune response, especially in cancer treatment.

[0009] A murine or humanized B7H3 antibody or a functional fragment thereof provided in the present disclosure includes a heavy chain sequence and a light chain sequence. For anti-human B7H3 murine antibodies, the amino acid sequence information of the heavy chain variable regions and the light chain variable regions is as follows: The amino acid sequences of 7F5's heavy chain variable region and light chain variable region are SEQ ID NO: 1 and 2, respectively; the amino acid sequences of 9C8's heavy chain variable region and light chain variable region are SEQ ID NO: 3 and 4, respectively; the amino acid sequences of 5B6's heavy chain variable region and light chain variable region are SEQ ID NO: 5 and 6, respectively; the amino acid sequences of 7C9's heavy chain variable region and light chain variable region are SEQ ID NO: 7 and 8, respectively; the amino acid sequences of 2A9's heavy chain variable region and light chain variable region are SEQ ID NO: 9 and 10, respectively; the amino acid sequences of 4F11's heavy chain variable region and light chain variable region are SEQ ID NO: 11 and 12, respectively; the amino acid sequences of 15A2's heavy chain variable region and light chain variable region are SEQ ID NO: 13 and 14, respectively; the amino acid sequences of 7B7's heavy chain variable region and light chain variable region are SEQ ID NO: 15 and 16, respectively; the amino acid sequences of 7E6's heavy chain variable region and light chain variable region are SEQ ID NO: 17 and 18, respectively; the amino acid sequences of 2E10's heavy chain variable region and light chain variable region are SEQ ID NO: 19 and 20, respectively; the amino acid sequences of 2F12's heavy chain variable region and light chain variable region are SEQ ID NO: 21 and 22, respectively; the amino acid sequences of 2F7's heavy chain variable region and light chain variable region are SEQ ID NO: 23 and 24, respectively; the amino acid sequences of 13A2's heavy chain variable region and light chain variable region are SEQ ID NO: 25 and 26, respectively; and the amino acid sequences of 14B3's heavy chain variable region and light chain variable region are SEQ ID NO: 27 and 28, respectively.

[0010] The amino acid sequence information of CDR1, CDR2, and CDR3 of the foregoing heavy chains and light chains of the antibodies is as follows: The amino acid sequences of 7F5's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 29, 30, and 31, respectively, and the amino acid sequences of 7F5's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 32, 33, and 34, respectively; the amino acid sequences of 9C8's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 35, 36, and 37, respectively, and the amino acid sequences of 9C8's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 38, 39, and 40, respectively; the amino acid sequences of 5B6's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 41, 42, and 43, respectively, and the amino acid sequences of 5B6's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 44, 45, and 46, respectively; the amino acid sequences of 7C9's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 47, 48, and 49, respectively, and the amino acid sequences of 7C9's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 50, 51, and 52, respectively; the amino acid sequences of 2A9's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 53, 54, and 55, respectively, and the amino acid sequences of 2A9's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 56, 57, and 58,

respectively; the amino acid sequences of 4F11's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 59, 60, and 61, respectively, and the amino acid sequences of 4F11's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 62, 63, and 64, respectively; the amino acid sequences of 15A2's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 65, 66, and 67, respectively, and the amino acid sequences of 15A2's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 68, 69, and 70, respectively; the amino acid sequences of 7B7's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 71, 72, and 73, respectively, and the amino acid sequences of 7B7's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 74, 75, and 76, respectively; the amino acid sequences of 7E6's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 77, 78, and 79, respectively, and the amino acid sequences of 7E6's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 80, 81, and 82, respectively; the amino acid sequences of 2E10's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 83, 84, and 85, respectively, and the amino acid sequences of 2E10's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 86, 87, and 88, respectively; the amino acid sequences of 2F12's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 89, 90, and 91, respectively, and the amino acid sequences of 2F12's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 92, 93, and 94, respectively; the amino acid sequences of 2F7's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 95, 96, and 97, respectively, and the amino acid sequences of 2F7's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 98, 99, and 100, respectively; the amino acid sequences of 13A2's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 101, 102, and 103, respectively, and the amino acid sequences of 13A2's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 104, 105, and 106, respectively; and the amino acid sequences of 14B3's heavy chain CDR1, CDR2, and CDR3 are SEQ ID NO: 107, 108, and 109, respectively, and the amino acid sequences of 14B3's light chain CDR1, CDR2, and CDR3 are SEQ ID NO: 110, 111, and 112, respectively. [0011] Further, the anti-human B7H3 antibody or fragment is modified to become a humanized antibody.

[0012] An amino acid sequence of a heavy chain variable region of anti-human B7H3 humanized antibody 2A9 is SEQ ID NO: 113, and an amino acid sequence of a light chain variable region of anti-human B7H3 humanized antibody 2A9 is SEQ ID NO: 114; an amino acid sequence of a heavy chain variable region of humanized antibody 4F11 is SEQ ID NO: 115, and an amino acid sequence of a light chain variable region of humanized antibody 4F11 is SEQ ID NO: 116; an amino acid sequence of a heavy chain variable region of humanized antibody 9C8 is SEQ ID NO: 117, and an amino acid sequence of a light chain variable region of humanized antibody 9C8 is SEQ ID NO: 118; an amino acid sequence of a heavy chain variable region of humanized antibody 15A2 is SEQ ID NO: 119, and an amino acid sequence of a light chain variable region of humanized antibody 15A2 is SEQ ID NO: 120; an amino acid sequence of a heavy chain variable region of humanized antibody 5B6 is SEQ ID NO: 121, and an amino acid sequence of a light chain variable region of humanized antibody 5B6 is SEQ ID NO: 122; and an amino acid sequence of a heavy chain variable region of humanized antibody 7B7 is SEQ ID NO: 123, and an amino acid sequence of a light chain variable region of humanized antibody 7B7 is SEQ ID NO: 124.

[0013] An expression vector includes a nucleic acid molecule of the antibody.

[0014] A pharmaceutical composition includes the antibody or a functional fragment thereof, and a pharmaceutical carrier.

[0015] Use of the antibody or the functional fragment, nucleic acid molecule, expression vector, host cell, and pharmaceutical composition thereof in preparing B7H3 immunomodulatory medications is provided.

[0016] In the present disclosure, recombinant B7H3 receptor proteins produced in a mammalian cell expression system are used as antigens to immunize mice, followed by fusing the mice's splenocytes with myeloma cells to obtain hybridoma cells. Multiple monoclonal hybridoma cell lines are obtained through extensive cloning and screening of these hybridoma cells. These hybridoma cell lines can secrete monoclonal antibodies (FIG. 1 and FIG. 2) that specifically bind to

B7H3 receptors. Some of these monoclonal antibodies have a significant effect of being endocytosed by cells (FIG. 3 and FIG. 4), some have an effect of promoting human immune cells to secrete cytokines such as IFN- γ (FIG. 5), and some have a significant effect of mediating ADCC (FIG. 6, FIG. 7 and FIG. 8). Some monoclonal antibodies are potential candidates for future development of ADC, ADCC, and B7H3 antagonists. The invention further involves cloning the genes encoding antibody light and heavy chain variable regions through RT-PCR (Reverse Transcription-Polymerase Chain Reaction), and constructing humanized antibodies using the complementarity-determining regions graft (CDR-graft) method. In vitro functional experiments show that the humanized B7H3 antibodies can specifically bind to B7H3 receptor proteins (FIG. 9 and FIG. 10) while maintaining significant cellular internalization (FIG. 11), have a significant cytotoxic effect on Calu-6 tumor cells post-MMAE toxin conjugation (FIG. 12), an ADCC mediation effect (FIG. 13), a mediation effect on ADCC cytotoxic effect of NK92MI-hCD16 cells on Jurkat-B7H3 and PC9 cells (FIG. 14), and an effect of promoting human immune cells to produce and secrete cytokines such as IFN- γ (FIG. 15). The positive control antibodies used in the present disclosure are derived from patent applications U.S. Pat. No. 8,802,091 and WO2017180813A1.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] In order to illustrate the technical solutions in the specific embodiments of the present disclosure or in the prior art more clearly, the following briefly describes the accompanying drawings used for describing the specific embodiments or the prior art. Apparently, the accompanying drawings in the following description show only some embodiments of the present disclosure, and those of ordinary skills in the art may derive other drawings from these accompanying drawings without creative efforts.

[0018] FIG. 1(a-b): Determination of the binding properties of B7H3 hybridoma monoclonal antibodies with B7H3-hFc protein using ELISA method;

[0019] FIG. 2(a-b): Determination of the binding properties of B7H3 hybridoma monoclonal antibodies with PC9 cells using FACS method;

[0020] FIG. 3(a-b): Determination of the cellular internalization activity of B7H3 hybridoma antibodies by Jurkat-B7H3 cells through FACS;

[0021] FIG. 4(a-b): Determination of the cellular internalization activity of B7H3 hybridoma antibodies by PC9 cells through FACS;

[0022] FIG. 5(a-b): Promoting effect of B7H3 hybridoma antibodies on IFN- γ secretion in human PBMC cells;

[0023] FIG. 6(a-b): Testing of the ADCC effect on Jurkat-B7H3 cells mediated by B7H3 hybridoma or chimeric antibodies using the reporter gene method;

[0024] FIG. 7(a-b): Testing of the ADCC effect on PC9 tumor cells mediated by some B7H3 chimeric antibodies using the reporter gene method;

[0025] FIG. 8: Testing of the ADCC effect on DLD1 tumor cells mediated by some B7H3 chimeric antibodies using the reporter gene method;

[0026] FIG. 9: Determination of the binding properties of humanized B7H3 monoclonal antibodies with B7H3-hFc protein using ELISA method;

[0027] FIG. 10: Determination of the binding properties of humanized B7H3 monoclonal antibodies with B7H3 protein in PC9 cells using FACS method;

[0028] FIG. 11: Determination of the cellular internalization effect of humanized B7H3 monoclonal antibodies by PC9 cells through FACS;

[0029] FIG. 12: Cytotoxic effect of humanized B7H3 antibody ADC on Calu-6 tumor cells;

[0030] FIG. 13(a-b): Testing of the ADCC effect of humanized B7H3 antibodies on Jurkat-B7H3 and PC9 cells using the reporter gene method;

[0031] FIG. 14(a-b): ADCC cytotoxic effect of NK92MI-hCD16 cells on Jurkat-B7H3 and PC9 cells mediated by humanized B7H3 antibodies; and

[0032] FIG. 15: Promoting effect of humanized B7H3 monoclonal antibodies on IFN- γ production and secretion by human PBMC cells.

DETAILED DESCRIPTION

[0033] As used herein, the term “antibody” is used in its broadest sense and includes immunoglobulin or other molecules containing one or more antigen-binding domains that specifically bind to an antigen, namely, a protein or peptide that exhibit binding specificity for a particular antigen. Specific examples of antibodies may include intact antibodies (for example, classic four-chain antibody molecules), single-chain antibodies, single-domain antibodies, bispecific antibodies, multispecific antibodies, and the like. A classic antibody molecule is usually a tetramer composed of two identical heavy chains and two identical light chains connected to each other through disulfide bonds. Based on a conservative difference in amino acid sequences, heavy and light chains are divided into variable regions (V) located at amino terminals and constant regions (C) located at carboxyl terminals. The variable region is used to recognize and bind the antigen, and the constant region (for example, Fc fragment) is used to initiate a downstream effect, for example, cell-mediated antibody-dependent cytotoxicity (ADCC) effect. Within the variable regions of the heavy chain and light chain, three local regions have a higher degree of variation in amino acid composition and sequences, are key binding positions for antibodies and antigens, and therefore, are also referred to as complementarity-determining regions (CDRs). The amino acid sequences of the CDRs can be readily determined by using numbering schemes recognized in the field, for example, Kabat, Chothia, IMGT, AbM or Contact.

[0034] An “antigen-binding fragment” of an antibody refers to an amino acid fragment in the antibody molecule that participates in specific antigen binding, for example, one of F(ab')₂, Fab and scFv.

[0035] EC₅₀ (concentration for 50% of maximal effect) refers to the concentration that causes 50% of the maximal effect. When used to indicate the binding ability of an antibody molecule to a corresponding antigen in an enzyme-linked immunosorbent assay (ELISA), EC₅₀ can refer to concentration of the antibody molecule when half of the maximum detection signal (for example, colorimetric or fluorescence intensity) occurs. The lower the value of EC₅₀, the greater the binding affinity to the antigen.

[0036] The present disclosure relates to an anti-B7H3 monoclonal antibody or an antigen-binding fragment thereof, including heavy chains and light chains, where an amino acid sequence of the heavy chain CDR1 is selected from one of SEQ ID NOs: 29, 35, 41, 47, 53, 59, 65, 71, 77, 83, 89, 95, 101 and 107; an amino acid sequence of heavy chain CDR2 is selected from one of SEQ ID NOs: 30, 36, 42, 48, 54, 60, 66, 72, 78, 84, 90, 96, 102 and 108; an amino acid sequence of heavy chain CDR3 is selected from one of SEQ ID NOs: 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91, 97, 103 and 109; an amino acid sequence of light chain CDR1 is selected from one of SEQ ID NOs: 32, 38, 44, 50, 56, 62, 68, 74, 80, 86, 92, 98, 104 and 110; an amino acid sequence of light chain CDR2 is selected from one of SEQ ID NOs: 33, 39, 45, 51, 57, 63, 69, 75, 81, 87, 93, 99, 105 and 111; and an amino acid sequence of light chain CDR3 is selected from one of SEQ ID NOs: 34, 40, 46, 52, 58, 64, 70, 76, 82, 88, 94, 100, 106 and 112, wherein the heavy and light chains of the antigen-binding fragment comprise amino acid sequences spanning from CDR1 to CDR3 of respective heavy and light chains of the antibody.

[0037] The anti-B7H3 antibody or antigen-binding fragment thereof provided in the present disclosure includes a heavy chain CDR and a light chain CDR, where the heavy chain CDR and the light chain CDR are selected from one or more groups of CDR combinations denoted as a to n:

TABLE-US-00001 Heavy Heavy Heavy Light Light Light Combination chain chain chain chain

chain chain name CDR1 CDR2 CDR3 a SEQ ID SEQ ID SEQ ID SEQ ID
SEQ ID SEQ ID NO: 29 NO: 30 NO: 31 NO: 32 NO: 33 NO: 34 b SEQ ID SEQ ID SEQ ID SEQ
ID SEQ ID SEQ ID NO: 35 NO: 36 NO: 37 NO: 38 NO: 39 NO: 40 c SEQ ID SEQ ID SEQ ID
SEQ ID SEQ ID SEQ ID NO: 41 NO: 42 NO: 43 NO: 44 NO: 45 NO: 46 d SEQ ID SEQ ID SEQ
ID SEQ ID SEQ ID SEQ ID NO: 47 NO: 48 NO: 49 NO: 50 NO: 51 NO: 52 e SEQ ID SEQ ID
SEQ ID SEQ ID SEQ ID SEQ ID NO: 53 NO: 54 NO: 55 NO: 56 NO: 57 NO: 58 f SEQ ID SEQ
ID SEQ ID SEQ ID SEQ ID SEQ ID NO: 59 NO: 60 NO: 61 NO: 62 NO: 63 NO: 64 g SEQ ID
SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID NO: 65 NO: 66 NO: 67 NO: 68 NO: 69 NO: 70 h SEQ
ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID NO: 71 NO: 72 NO: 73 NO: 74 NO: 75 NO: 76 i
SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID NO: 77 NO: 78 NO: 79 NO: 80 NO: 81 NO: 82
g SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID NO: 83 NO: 84 NO: 85 NO: 86 NO: 87 NO:
88 k SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID NO: 89 NO: 90 NO: 91 NO: 92 NO: 93
NO: 94 l SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID NO: 95 NO: 96 NO: 97 NO: 98 NO:
99 NO: 100 m SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID NO: 101 NO: 102 NO: 103 NO:
104 NO: 105 NO: 106 n SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID SEQ ID NO: 107 NO: 108 NO:
109 NO: 110 NO: 111 NO: 112

[0038] In some embodiments, an amino acid sequence of a heavy chain variable region is selected from one of SEQ ID NOs: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25 and 27, and an amino acid sequence of a light chain variable region is selected from one of SEQ ID NOs: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26 and 28.

[0039] In some embodiments, the anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof includes a heavy chain variable region and a light chain variable region, wherein the heavy chain variable region and the light chain variable region are selected from one or more groups of variable region combinations denoted as A to N:

TABLE-US-00002 Combination Heavy chain Light chain name variable region variable region A
SEQ ID NO: 1 SEQ ID NO: 2 B SEQ ID NO: 3 SEQ ID NO: 4 C SEQ ID NO: 5 SEQ ID NO: 6 D
SEQ ID NO: 7 SEQ ID NO: 8 E SEQ ID NO: 9 SEQ ID NO: 10 F SEQ ID NO: 11 SEQ ID NO: 12
G SEQ ID NO: 13 SEQ ID NO: 14 H SEQ ID NO: 15 SEQ ID NO: 16 I SEQ ID NO: 17 SEQ ID
NO: 18 J SEQ ID NO: 19 SEQ ID NO: 20 K SEQ ID NO: 21 SEQ ID NO: 22 L SEQ ID NO: 23
SEQ ID NO: 24 M SEQ ID NO: 25 SEQ ID NO: 26 N SEQ ID NO: 27 SEQ ID NO: 28

[0040] Based on general understandings of persons skilled in the art, functional variants of the antibody or its antigen-binding fragment are also encompassed within the scope of this disclosure. The term “functional variant” used herein refers to a variant molecule obtained by introducing one or more amino acids for insertion, deletion or substitution based on the parent protein molecule (for example, a native protein molecule), where the variant molecule still retains at least some of the functions of the parent protein molecule (especially a function of interest, for example, an ability to bind to the corresponding antigen).

[0041] For the CDR region, the functional variants of the present disclosure include a heavy chain complementarity-determining region and a light chain complementarity-determining region. Compared with any one of the complementation-determining region groups denoted as a to n, the heavy chain complementarity-determining region and the light chain complementarity-determining region each contain mutations of up to 3 amino acids (for example, substitution, deletion or addition of 1, 2 or 3 amino acids or any combination thereof). Preferably, the mutations are conservative mutations.

[0042] For the variable region, in some embodiments, the functional variant in the present disclosure includes a heavy chain variable region, and an amino acid sequence of the heavy chain variable region includes a sequence sharing at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identity with any one of the heavy chain variable region sequences shown in the groups A to N. In some embodiments, the functional variant in the present disclosure includes a light chain variable region, and an amino acid sequence of the light chain variable region

includes a sequence sharing at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identity with any one of the light chain variable region sequences shown in the groups A to N. The term “identity” refers to a quantity of a degree of identity between two amino acid or nucleotide sequences (for example, a query sequence and a reference sequence), and is generally expressed as a percentage. Usually, before the percentage of identity between two amino acid or nucleotide sequences is calculated, the two amino acid or nucleotide sequences are first subjected to sequence alignment and a gap (if any) is introduced. At a certain alignment position, if amino acid residues or bases in the two sequences are the same, the two sequences are consistent or matched at the position; or if the amino acid residues or bases in the two sequences are different, the two sequences are considered to be inconsistent or mismatched at the position. In some algorithms, the number of matched positions is divided by the total number of positions in the alignment window to obtain the sequence identity. In other algorithms, the number of gaps and/or gap length is also taken into account. Commonly used sequence alignment algorithms or software include DANMAN, CLUSTALW, MAFFT, BLAST, MUSCLE, among others. For the purposes of the present disclosure, the publicly available alignment software BLAST (available at <https://www.ncbi.nlm.nih.gov/>) can be used to obtain optimal sequence alignment through default settings and calculate the sequence identity between two amino acid or nucleotide sequences.

[0043] The functional variant retains the ability to specifically bind B7H3. Persons skilled in the art can determine a suitable variant of the antigen-binding molecule illustrated herein by using well-known technologies. In certain embodiments, persons skilled in the art can identify regions in the antibody or antigen-binding fragment thereof that are unimportant for the activity of specifically binding B7H3 and change suitable regions without destroying the activity.

[0044] In some embodiments, the B7H3 monoclonal antibody or antigen-binding fragment thereof is murine or humanized.

[0045] In some embodiments, the heavy chain variable region and light chain variable region are humanized.

[0046] In some embodiments, the heavy and light chains are humanized.

[0047] The term “humanized antibody”, also known as CDR-grafted antibody, refers to an antibody produced by transplanting the first animal-derived CDR sequence into human antibody variable region frameworks, that is, different types of human germline antibody framework sequences. Humanized antibodies can overcome the heterologous reactions induced by chimeric antibodies due to their significant murine protein content. Such framework sequences can be obtained from public DNA databases or published references containing germline antibody gene sequences. For example, germline DNA sequences of human heavy and light chain variable region genes are available in “VBase” human germline sequence database (www.mrccpe.com.ac.uk/vbase) and found in Sequences of Proteins of Immunological Interest (Kabat, E. A. et al., 1991, 5.sup.th ed.) To avoid both a decrease in immunogenicity and a resulting decrease in activity, a minimum of reverse mutation or back mutation can be performed on the human antibody variable region framework sequence to maintain activity. The humanized antibody in the present disclosure also includes a mature humanized antibody obtained after phage shows affinity to the CDR. In a preferred embodiment of the present disclosure, the first animal origin is of murine origin.

[0048] In some embodiments, the heavy chain and the light chain are humanized; an amino acid sequence of a humanized heavy chain variable region is selected from one of SEQ ID NOs: 113, 115, 117, 119, 121, and 123; and the amino acid sequence of the humanized light chain variable region is selected from one of SEQ ID NOs: 114, 116, 118, 120, 122, and 124.

[0049] In some embodiments, the humanized heavy chain variable region and light chain variable region are selected from one or more groups of variable region combinations denoted as 1 to 6:

TABLE-US-00003	Combination	Heavy chain	Light chain	name	variable region	variable region
1	SEQ ID NO: 113	SEQ ID NO: 114	2	SEQ ID NO: 115	SEQ ID NO: 116	3
2	SEQ ID NO: 117	SEQ ID NO: 118	4	SEQ ID NO: 119	SEQ ID NO: 120	5
3	SEQ ID NO: 121	SEQ ID NO: 122	6	SEQ ID NO: 123	SEQ ID NO: 124	

NO: 123 SEQ ID NO: 124

[0050] In some embodiments, the heavy chain and the light chain are humanized; an amino acid sequence of a humanized heavy chain variable region is an amino acid sequence with SEQ ID NO: 113; and the amino acid sequence of the humanized light chain variable region is an amino acid sequence with SEQ ID NO: 114.

[0051] In some embodiments, the anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof includes a constant region of human IgG1.

[0052] The present disclosure further relates to use of the anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof in preparation of the following medication: a medication for regulating B7H3 activity or B7H3 level, a medication that has significant cellular internalization (allowing selective tumor killing or inhibition through toxin conjugation), significant antibody-dependent cytotoxicity (ADCC), or an ability to block B7H3 in immunosuppression to enhance body immunity, a medication for promoting T lymphocyte, or a medication for improving production of cytokine such as IFN- γ in T lymphocyte.

[0053] The present disclosure further relates to a monoclonal antibody conjugate, including a monoclonal antibody and a conjugating part, wherein the monoclonal antibody is one of the previously mentioned anti-B7H3 monoclonal antibodies or their antigen-binding fragments, and the conjugating part is selected from one or more of a radionuclide, a medication, a toxin, a cytokine, a cytokine receptor fragment, an enzyme, fluorescein, and biotin.

[0054] The present disclosure further relates to use of the monoclonal antibody conjugate in preparation of the following medication: a medication for regulating B7H3 activity or B7H3 level, a medication having significant cellular internalization (allowing selective tumor killing or inhibition through toxin conjugation), a medication having significant antibody-dependent cytotoxicity (ADCC), a medication for blocking B7H3 in immunosuppression to enhance body immunity, a medication for promoting T lymphocyte, or a medication for improving production of cytokine such as IFN- γ in T lymphocyte.

[0055] The present disclosure further relates to use of the foregoing monoclonal antibody conjugate in preparation of a medication for preventing, treating, and/or adjuvantly treating a tumor.

[0056] The present disclosure further relates to an expression vector, including a nucleic acid molecule of the antibody.

[0057] The present disclosure further relates to an expression vector, including a nucleic acid molecule of the foregoing anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof.

[0058] The term “vector” refers to a nucleic acid delivery vehicle into which a polynucleotide can be inserted. When a vector enables an inserted polynucleotide-encoded protein to be expressed, the vector is referred to as an expression vector. The vector can enter a host cell via transformation, transduction, or transfection, so that an element of a genetic material carried by the vector is expressed in the host cell. The vector is well-known to a person skilled in the art and includes, but is not limited to, a plasmid, a phagemid, a cosmid, an artificial chromosome (for example, a yeast artificial chromosome (YAC), a bacterial artificial chromosome (BAC), or a P1 derived artificial chromosome (PAC)), a phage (λ phage or M13 phage), and an animal virus. An animal virus that can serve as the vector includes, but is not limited to, retrovirus (including lentivirus), adenovirus, adeno-associated virus, herpesvirus (for example, herpes simplex virus), poxvirus, baculovirus, papillomavirus, and papillomavirus (for example, SV40). In some embodiments, the vector in the present disclosure includes a regulatory element commonly used in genetic engineering, for example, the enhancer, the promoter, an internal ribosome entry site (IRES) and another expression control element (for example, a transcription termination signal, or a polyadenylation signal and a polyU sequence).

[0059] In the present disclosure, the vector can be a composition, for example, a mixture of multiple plasmids, where different plasmids carry part of the antibody or the antigen-binding fragment.

[0060] The present disclosure further relates to a pharmaceutical composition, including the foregoing antibody or a (at least one) functional fragment thereof, and a pharmaceutical carrier. [0061] As used herein, the “pharmaceutical carrier” includes any material that allows a component to remain biologically active when combined with the active component and that does not react with an immune system of the subject. An example includes, but is not limited to, any one of a standard drug carrier (for example, a phosphate buffered saline solution, water, an emulsion (for example, an oil/water emulsion)) and various types of wetting agents. A composition containing such carrier is prepared in a well-known conventional method (refer to, for example, Remington's Pharmaceutical Sciences, 18th edition, A. Gennaro, ed., Mack Publishing Co., Easton, PA, 1990; and Remington, The Science and Practice of Pharmacy, 21st edition, Mack Publishing, 2005). [0062] The present disclosure further relates to use of the foregoing antibody or the functional fragment, nucleic acid molecule, expression vector, host cell, and pharmaceutical composition thereof in preparing B7H3 immunomodulatory medications.

[0063] The embodiments of the present disclosure are described in detail below with reference to examples:

Example 1

Immunization and Cell Fusion in Mice for Producing Anti-B7H3 Antibodies

[0064] Human B7H3 (NCBI Reference Sequence: NM_001024736.2) extracellular domain and mFc fusion protein (B7H3-ECD-mFc) were used as antigens, and fully emulsified with an equal volume of complete Freund's adjuvant (Sigma, Cat. No.: F5581), 6-8 week old Balb/c mice (Zhaoyan (Suzhou) New Drug Research Center Co., Ltd.) were subcutaneously immunized with a dose of 50 µg antigen per mouse. Subsequently, the mice were immunized subcutaneously three times with the same dose of antigen fully emulsified in incomplete Freund's adjuvant (Sigma, Cat. No.: F5506) every 2 weeks. After three immunizations, serum titers of the mice were measured, and the mice were booster immunized once via intraperitoneal injection 3 days before the fusion. Using PEG Hybri-Max (Sigma, Cat.: 7181) as a fusion agent, splenocytes of the mice and SP2/0 cells were mixed in a ratio of 4:1. Fused cells were added into a 96-well plate (1×10⁵ cells/well), and each well contained 0.1 mL of 1×HAT medium (Invitrogen, Cat. No.: 21060-017). 0.1 mL of HT (Invitrogen, Cat. No.: 11067-030) medium was added on day 3, the medium was pipetted out of the 96-well plate on day 7, and 0.2 mL of fresh HT medium was added. The supernatant was collected on day 9 for various screening and tests.

Example 2

Antigen Binding Ability of Hybridoma Antibodies and Subcloning

[0065] The binding ability of hybridoma antibodies was tested using ELISA and FACS, along with subcloning via the limiting dilution method:

(1) Screening of B7H3-binding positive clones through ELISA: 50 µL of B7H3-hFc (final concentration: 2 µg/mL) was used to coat each well of a 96-well ELISA plate (Corning, Cat. No.: 9018), and incubated overnight at room temperature. After the ELISA plate was rinsed with washing buffer (PBS+0.05% Tween20) 3 times, blocking buffer (PBS+2% BSA (Sigma, Cat. No.: V90093)) was added and the ELISA plate was incubated at room temperature for 1 hour. The ELISA plate was rinsed with washing buffer 3 times; and hybridoma supernatant was added, and the ELISA plate was incubated at room temperature for 1 hour and rinsed 3 times. 100 µL of 10,000-fold diluted HRP-conjugated goat anti-mouse IgG secondary antibody (Thermo, Cat. No.: 31432) was added to each well, and the ELISA plate was incubated at room temperature for 1 hour in the dark and rinsed 3 times. 100 µL of TMB (Biocytogen, Cat. No.: ES-002) was added to each well, and the ELISA plate was incubated at room temperature for color development for 2 minutes, 100 µL of reaction terminating solution (2N H₂SO₄) was added to each well to terminate the color reaction, and an OD₄₅₀ value of each well was read using a microplate reader (Tecan Spark).

(2) Screening of B7H3-binding positive clones through FACS: 50 µL of the positive hybridoma supernatant or purified hybridoma antibody in the foregoing test was mixed with 50 µL of 293T-

B7H3 cells (2×10⁵ cells/well), and a resulting mixture was added to a 96-well plate with a U-shaped bottom and left standing at 4° C. for 1 hour, the 96-well plate was rinsed with FACS buffer (PBS+3% FCS) and centrifuged twice, then 400-fold diluted PE-labeled goat anti-mouse secondary antibody (Biolegend, Cat. No.: 405307) was added, the 96-well plate was incubated at 4° C. for 30 minutes in the dark, rinsed with FACS buffer and centrifuged twice, and signal values of the cells in a PE channel were detected by using BD Accuri C6 flow cytometer.

(3) Subcloning: Positive multiclonal hybridoma cells identified by ELISA and FACS were subcloned using the limiting dilution method. The subclones were further detected and screened using ELISA and FACS to obtain positive hybridoma monoclonal clones.

Antibody Purification, Concentration Determination, and Endotoxin Testing:

[0066] The positive monoclonal hybridoma cells were added to 50 ml of serum-free medium (Invitrogen, Cat. No.: 12045-076) and the serum-free medium was cultured for 8 to 9 days, and then centrifuged to collect supernatant. The monoclonal antibodies were purified through protein A affinity chromatography. The purified antibody samples were liquid-exchanged and concentrated in ultrafiltration centrifuge tubes (Millipore, Cat. No.: ACS500024). Protein concentration was determined in a BCA method and limulus reagent (Xiamen Bioendo Technology Co., Ltd.) was used to detect endotoxin content of purified antibody samples.

[0067] ELISA and FACS were conducted to detect the binding ability of purified antibody samples to B7H3. Results are shown in FIG. 1(a-b), FIG. 2(a-b) and Table 1. The selected hybridoma antibodies had higher affinity.

TABLE-US-00004 TABLE 1 EC.sub.50 values for the binding of B7H3 hybridoma monoclonal antibodies to the B7H3 antigen

Antibody	Parameter	7F5	9C8	5B6	7C9	2A9	4F11	15A2	7B7	7E6	2E10	2F12	2F7	13A2	14B3
ELISA EC.sub.50		5.76	10.45	4.93	9.9	6.24	5.11	1.51	9.7	4.49	6.7	na	2.78	0.52	9.1
(ng/mL)															
FACS EC.sub.50		197	86.1	71.3	103.5	186.5	83.4	6.2	179.2	52.6	234	217	177	100.7	248.5
(ng/mL)															

Example 3

Testing Intracellular Phagocytosis and B7H3 Blocking Action of Hybridoma Monoclonal Antibodies

[0068] Analysis of intracellular phagocytosis of antibodies: In a 96-well plate, 50 μL of 2×10⁵/mL PC9 cells and 50 μL of labeled B7H3 antibodies were added to each well, and the 96-well cell culture plate was incubated at 4° C. for 1 hour. After the cell culture plate was rinsed with FACS buffer twice, Goat anti-mFc (Jackson, Cat. No.: 115-005-071) secondary antibody labeled with pH-dependent fluorescent dye CypHer5E (GE, Cat. No.: PA15401) was added, and the cell culture plate was incubated at 4° C. for 0.5 hours and rinsed twice. A culture medium (1640+10% FBS) was added, the cell culture plate was put in an incubator and incubated for 3 hours, and then centrifuged to remove the culture medium. A residue was resuspended in PBS at pH 9.0, and CypHer5E signal of PC9 cells was detected through a BD C6 flow cytometer.

Efficiency of entering the cells by B7H3 antibodies was calculated. As shown in FIG. 3(a-d) and FIG. 4(a-d), Jurkat-B7H3 and PC9 cells had significant endocytic activity for B7H3 antibodies, including 15A2, 5B6, 7C9, 2F7, 7B7, 2E10 and 4F11.

[0069] Testing of the effect of B7H3 hybridoma antibodies on secretion of cytokines by human PBMC cells: PBMC cells (TPCS, Cat. No.: PB025C) resuspended in complete medium (RPMI1640+10% FCS) was added into the 96-well plate (Corning, Cat. No.: 3799), then 40 ng/ml OKT3 (eBioscience, Cat. No.: 16-0037-85) was added, and the 96-well plate was incubated at 37° C. for 72 hours. After counting, the activated PBMC cells were resuspended in complete culture medium (2.5×10⁵ cells/mL). In the 96-well plate, 100 μL of PBMC cells and 50 μL of B7H3 antibodies at different concentrations (initial concentration of 20 μg/mL, 10-fold serial dilution) were added to each well, the 96-well cell culture plate (Corning, Cat. No.: 3599) was placed in an incubator containing 5% CO₂ at 37° C. for 48 hours, and the supernatant was collected. The concentrations of cytokines were detected using IFN-γ ELISA kit (R&D Systems, Cat. No.:

DY285). As shown in FIG. 5, B7H3 antibodies including 5B6, 2F12, 2F7, 15A2, 13A2, 14B3, 4F11, 2A9, etc., can significantly promote the secretion of IFN- γ by PBMC cells.

Example 4

Cloning and Sequencing of B7H3 Antibody Variable Region Genes

[0070] B7H3 monoclonal hybridoma cell lines were lysed with TRIzol (Cwbiotech, Cat No: CW0580), and the total RNAs of hybridoma cells were extracted. RNA of hybridoma cells was reversely transcribed into cDNA by using HiFi Script cDNA Synthesis Kit (Cwbiotech, Cat No: CW2569). The cDNA was used as a template, degenerate primers were used to amplify heavy chain and light chain variable region genes of the antibody in a PCR method (Kettleborough et al. (1993) Eur. J Immunology 23:206-211; Strebe et al. (2010) Antibody Engineering 1:3-14). After the product of the PCR amplification was ligated to T/A vector, DH5 α competent cells were transformed, inoculated and cultured overnight at 37° C. Monoclonal was picked from the culture plate and cultured for amplification, plasmids were extracted, and the gene sequence of the antibody was identified. Based on the gene sequence of the antibody, complementarity-determining regions (CDRs) and framework regions of the antibody were analyzed. The B7H3 antibody sequence number is shown in Table 2, and specific sequence information is shown in the sequence listing.

TABLE-US-00005	TABLE 2	B7H3 Antibody	Sequence	Number	SEQ ID	Description	NO.
Antibody Sequence	description	Amino acid	sequence	1	7F5	Amino acid sequence of a	
EVQLQQSGAELVRPGALVKLSCKTSGF	heavy chain variable region						
NIKAYYIHWVKQRPEQGLEWIGWIDPE	NGNTFYDPKFQGKASITADTSSNTAYLQ						
LSSLTSEDAVYYCTRSSTRLRPFDYW	QGGTTLTVSS	2	7F5	Amino acid	sequence of a		
DIVMTQAAFSNPVTLTGTSASISCRSSKS	light chain variable region						
LLHSNGITYLYWYLQKSGQSPQLLIYQM	SNLASGVPDRFSSSGSGTDFTLRISRV						
EAEDVG VYYCAQNLELYTFGGG	TKLEIK	3	9C8	Amino acid	sequence of a		
EVQLQQSGPELVKPGASVKISCKTSGY	heavy chain variable region						
TFTEYTIHWVKQSHGKSLEWIGGINPNS	GGTTYNLKFKGKATLTVDKSSSTVYME						
LRSLTSEDSAVYYCARKEGFFTDYWG	QGGTTLTVSS	4	9C8	Amino acid	sequence of a		
QTVLTQSPAIMSASPGEKVTMTCSASS	light chain variable region						
SLSYIHWYRQKSGTSPKRWIYDTSNLA	TGVPVRFSGSGSGTSYSLTISSMEAED						
AATYYCQQWHYNPPTFGGG	TKLEIK	5	5B6	Amino acid	sequence of a		
QVQLQQSGAELARPGASVKMSCKTSG	heavy chain variable region						
YTFITYTIHWVKQRPGQGLEWIGYINPT	SDYANFNQKFQDKATLTADKSSRTAYM						
QLSSLTSEDSAVYYCARWGTTVFQFYF	DYWGQGGTTLTVSS	6	5B6	Amino acid	sequence of a		
QIVLSQSPAILSASPGEKVTMTCRASQS	light chain variable region						
LSYINWYQLKPGSSPKPWYATSSLQS	GVPARFSGSASGTSYSLTISRVEAEDAA						
TYFCQHWSSDPPTIGGG	TKLEIK	7	7C9	Amino acid	sequence of a		
EVQLQQSGTELVRPGALVKLSCKASAL	heavy chain variable region						
NIKDFYMHWVKQRPERGLEWIGWIDPD	NGNTIYDPKFQGKASITADISSSTAYLQL						
SSLTSED TAVYYCARSTRQRPFDYWG	QGGTTLTVSS	8	7C9	Amino acid	sequence of a		
DIVMTQSHKVMSTSVGDRVSITCKASQ	light chain variable region						
DVRTGVAWYQQKPGQSPKLLIYSAFYR	YTGVDPDRFTVSGSGADFTFTITSVQAE						
DLAVYYCQQYYSIPWTFGGG	TKLEIK	9	2A9	Amino acid	sequence of a		
EVQLQQSGPELVKPGASVKISCKTSGY	heavy chain variable region						
TFTEYIMHWVKQSRGKSLEWIGGINPN	SGGTTYNEKFKGKATLTVDKSASTAYM						
ELRSLTSEDSAIYYCARRMPYWFYFDVW	GAGTTVT TVSS	10	2A9	Amino acid	sequence of a		
QIVLTQSPAIMSASPGEKVTMTCSASSS	light chain variable region						
VSYIHWYRQKSGTSPKRWVYDTSKLAS	GVPARFSGSGSGTSYSLTISSMEAEDA						
ATYYCQQWVSNPLTFGAG	TKLELK	11	4F11	Amino acid	sequence of a		
EVQLQQSGPELVKPGASVKLSCKTSGY	heavy chain variable region						

TFTEYIMHWKQSHAKSLEWIGGINPN TGGTINQKFKDKATLTVDKSSSTAYME
LRSLTSDDSAVYFSARAGGNFLWYFDV WGAGTTVTVSS 12 4F11 Amino acid
sequence of a QIVLTQSPATMSASPGEKVTMTCSASS light chain variable region
SVSYMHWYQQKSDMSPKRWLYDTSKL ASGVPARFSGSGSGTSYSLAISSMEAE
DAATYYCQQWNSNPLTFGTGTKLELK 13 15A2 Amino acid sequence of a
QVQLQQSGPEVVKPGALVRISCKASGY heavy chain variable region
TFTSYDINWVKQRPGQGLEWIGWIDPG DGSTKYNEKFKGKATLTADKSSSTAYM
QFSSLTSENSAVYFCVRRGFDYWGGQ TTLTVSS 14 15A2 Amino acid sequence of a
QILLTQSPAISASPGEKVTMTCRTSSS light chain variable region
VSYLHWFQQKPGSSPKLWIHDTSNLAS GFPARFSGSGSGTSYSLIISMEAEADA
TYYCHQRSSYPWTFGGGTGLEIK 15 7B7 Amino acid sequence of a
EVKLVESGGGLVLRPGGSLKLSAASGF heavy chain variable region
TFSSYAVSWVRQTPEKRLEWVASISGG GIYIYYPDSVKGRFTISRDNKNNLYLQ
MNSLTPDDTALYYCARHGGAGYFDYW GQGTTTLTVSS 16 7B7 Amino acid sequence
of a DSLMTQSPASLSASVGETVTITCRGSE light chain variable region
SVHSYLAWYQQKQKSPQLLVYNAKTL QSGVPSRFSGSGSGTQFSLRINSLQPE
DFGTYYCQQHYGSPPWTFGGGTILEIK 17 7E6 Amino acid sequence of a
QVQLQQSGPELVRPGVSVKISCKGSGY heavy chain variable region
TFTDYAMHWVKQSHAKSLEWIGVINSY YGNTHYNQKFKGKATMTVDKSSSTAYM
ELARLTSEDSAIYYCARAGGYSYGSSY RPDRFAYWGQGLVIVSA 18 7E6 Amino acid
sequence of a DIQMTQSPASQSASLGESVTITCLASQT light chain variable region
IGTWLAWYQQKPGKSPQLLIYAATNLAD GVPSRFSGSGSGTKFSFKISSLQAEDF
VSYYCQQLYSTPYTFGGGTGLEIK 19 2E10 Amino acid sequence of a
EVQLQQSGPELVKPGASVKISCKTSGY heavy chain variable region
TFTEYTMHWVKQSHGKNLEWIGGINPD NGGTTYSQKFKDKATLTVDKSSSTAYM
VLRSLTSEDSAVYYCARSFNNYGPWF AYWGQGLVTVSA 20 2E10 Amino acid
sequence of a DIVMTQSHKFMSTSIGDRVSITCKASQD light chain variable region
VRSAAWYQQRPGQSPKLLIYWASTRH TGVPARFTGSGSGTDYTLTINNVQSED
LAGYFCQQYSSYFTFGGGTKLEKK 21 2F12 Amino acid sequence of a
EVQLQQSGPELVKSGASVNISCKTSGY heavy chain variable region
TFTENTMHWVKLSHGKSLEWIGGINPN NGGTTYNQKFKGKATLTVDKSSSTAYM
ELRSLTSEDSAVYYCARRGRWPPDWY FDVWGAGTTVTVSS 22 2F12 Amino acid
sequence of a QIVLTQSPAISASPGEKVTMTCSASSS light chain variable region
LSYMHWYQQKSGTSPKRWIYDTSKLA SGVPARFSGSGSGTSYSLTISSMEAE
AATYYCQQWSSNPLTFGAGTKLELK 23 2F7 Amino acid sequence of a
QVQLQQPGAELVKPGASVKLSCKASG heavy chain variable region
YTFTSYIYWLKQRPGQGLEWIGGIKPS NGDTNFMNAKFMNKATLTVDKSSSTAYM
QLSSLTSEDSAIYYCTRFFYGYSPYDFDV WGAGTTVTVSS 24 2F7 Amino acid sequence
of a NIVMTQSPKSMMAVGERVTLTCKASE light chain variable region
NVDTHVSWYQQKPEQSPTLLMYGTSN RFTGVPDRFTGSGSATDFTLTISVQAE
DLADYHCGQSYRYPYTFGGGTGLEIK 25 13A2 Amino acid sequence of a
EVQLQQSGPELVKPGASVKISCKTSGY heavy chain variable region
TFTEYVMHWVKQRHGQSLEWIGGINP NTGGTTYNQKFEGKATLTVDKSSSSAY
MEFRSLTSEDSAVYYCARRMPFGISAM DYWGQGTSTVTVSS 26 13A2 Amino acid
sequence of a QIVLTQSPAISASPGEKVTMTCSASSS light chain variable region
VSYIHWYQQKSGTSPKRWISETSKLA SGVPARFSGSGSGTSYSLTISSMGAEDA
ATYYCHQWISYPPTFGAGTKLELK 27 14B3 Amino acid sequence of a
QVQLQQSGPELVRPGVSVKISCKGSGY heavy chain variable region
TFTDHAIHWVKQSHAKSLEWIGVINTYS GNTNFMNQKLKGKATMTVDKSSSTAYME
LARLTSEDSAIYYCARGAHSSGYPYWY FDVWGAGTTVTVSS 28 14B3 Amino acid

sequence of a QIVLSQSPAILSTASAGEVTMTCRASSS light chain variable region
VSYMHWYQQKPGSSPKPWIYATSNLA SGVPARFSGSGSGTSYSLTISRVEAED
AATYYCQQWSTNPFTEGSGTKLEMK 29 7F5 Amino acid sequence of a AYYIH
heavy chain CDR1 region 30 7F5 Amino acid sequence of a
WIDPENGNTFYDPKFQG heavy chain CDR2 region 31 7F5 Amino acid sequence
of a SSTRLRPFDY heavy chain CDR3 region 32 7F5 Amino acid sequence of a
RSSKSLLSNGITYLY light chain CDR1 region 33 7F5 Amino acid sequence of a
QMSNLAS light chain CDR2 region 34 7F5 Amino acid sequence of a AQNLELYT
light chain CDR3 region 35 9C8 Amino acid sequence of a EYTIH heavy chain
CDR1 region 36 9C8 Amino acid sequence of a GINPNSGGTTYNLKFKG heavy
chain CDR2 region 37 9C8 Amino acid sequence of a KEGFFTDDY heavy chain
CDR3 region 38 9C8 Amino acid sequence of a SASSLSYIH light chain CDR1
region 39 9C8 Amino acid sequence of a DTSNLAT light chain CDR2 region 40 9C8
Amino acid sequence of a QQWHYNPPT light chain CDR3 region 41 5B6 Amino
acid sequence of a TYTIH heavy chain CDR1 region 42 5B6 Amino acid
sequence of a YINPTSDYANFNQKFKD heavy chain CDR2 region 43 5B6 Amino
acid sequence of a WGTTFVFQFYFDY heavy chain CDR3 region 44 5B6 Amino
acid sequence of a RASQSLSYIN light chain CDR1 region 45 5B6 Amino acid
sequence of a ATSSLQS light chain CDR2 region 46 5B6 Amino acid sequence
of a QHWSSDPPT light chain CDR3 region 47 7C9 Amino acid sequence of a
DFYMH heavy chain CDR1 region 48 7C9 Amino acid sequence of a
WIDPDNGNTIYDPKFQG heavy chain CDR2 region 49 7C9 Amino acid sequence
of a SSTRQRPFDY heavy chain CDR3 region 50 7C9 Amino acid sequence of a
KASQDVRTGVA light chain CDR1 region 51 7C9 Amino acid sequence of a
SAFYRYT light chain CDR2 region 52 7C9 Amino acid sequence of a QQYYSSIPWT
light chain CDR3 region 53 2A9 Amino acid sequence of a EYIMH heavy chain
CDR1 region 54 2A9 Amino acid sequence of a GINPNSGGTTYNEKFKG heavy
chain CDR2 region 55 2A9 Amino acid sequence of a RMPYWYFDV heavy chain
CDR3 region 56 2A9 Amino acid sequence of a SASSSVSYIH light chain CDR1
region 57 2A9 Amino acid sequence of a DTSKLAS light chain CDR2 region 58 2A9
Amino acid sequence of a QQWVSNPLT light chain CDR3 region 59 4F11 Amino
acid sequence of a EYIMH heavy chain CDR1 region 60 4F11 Amino acid
sequence of a GINPNTGGITYNQKFKD heavy chain CDR2 region 61 4F11 Amino
acid sequence of a AGGNFLWYFDV heavy chain CDR3 region 62 4F11 Amino
acid sequence of a SASSSVSYMH light chain CDR1 region 63 4F11 Amino acid
sequence of a DTSKLAS light chain CDR2 region 64 4F11 Amino acid sequence
of a QQWNSNPLT light chain CDR3 region 65 15A2 Amino acid sequence of a
SYDIN heavy chain CDR1 region 66 15A2 Amino acid sequence of a
WIDPGDGSTKYNEKFKG heavy chain CDR2 region 67 15A2 Amino acid sequence
of a RGFDY heavy chain CDR3 region 68 15A2 Amino acid sequence of a
RTSSSVSYLH light chain CDR1 region 69 15A2 Amino acid sequence of a
DTSNLAS light chain CDR2 region 70 15A2 Amino acid sequence of a
HQRSSYPWT light chain CDR3 region 71 7B7 Amino acid sequence of a SYAVS
heavy chain CDR1 region 72 7B7 Amino acid sequence of a
SISGGGIYIYYPDSVKG heavy chain CDR2 region 73 7B7 Amino acid sequence of
a HGGAGYFDY heavy chain CDR3 region 74 7B7 Amino acid sequence of a
RGSESVHSYLA light chain CDR1 region 75 7B7 Amino acid sequence of a
NAKTLQS light chain CDR2 region 76 7B7 Amino acid sequence of a
QQHYGSPPWT light chain CDR3 region 77 7E6 Amino acid sequence of a DYAMH
heavy chain CDR1 region 78 7E6 Amino acid sequence of a

VINSYYGNTHYNQKFKG heavy chain CDR2 region 79 7E6 Amino acid sequence of a AGGYSYGSSYRPDRFAY heavy chain CDR3 region 80 7E6 Amino acid sequence of a LASQTIGTWLA light chain CDR1 region 81 7E6 Amino acid sequence of a AATNLAD light chain CDR2 region 82 7E6 Amino acid sequence of a QQLYSTPYT light chain CDR3 region 83 2E10 Amino acid sequence of a EYTMH heavy chain CDR1 region 84 2E10 Amino acid sequence of a GINTDNGGTTYSQKFKD heavy chain CDR2 region 85 2E10 Amino acid sequence of a SFNNYGPAWFAY heavy chain CDR3 region 86 2E10 Amino acid sequence of a KASQDVRSAVA light chain CDR1 region 87 2E10 Amino acid sequence of a WASTRHT light chain CDR2 region 88 2E10 Amino acid sequence of a QQYSSYFT light chain CDR3 region 89 2F12 Amino acid sequence of a ENTMH heavy chain CDR1 region 90 2F12 Amino acid sequence of a GINPNNGGTTYNQKFKG heavy chain CDR2 region 91 2F12 Amino acid sequence of a RGRWPPDWYFDV heavy chain CDR3 region 92 2F12 Amino acid sequence of a SASSSLSYMHL light chain CDR1 region 93 2F12 Amino acid sequence of a DTSKLAS light chain CDR2 region 94 2F12 Amino acid sequence of a QQWSSNPLT light chain CDR3 region 95 2F7 Amino acid sequence of a SYIY heavy chain CDR1 region 96 2F7 Amino acid sequence of a GIKPSNGDTNFNKFMN heavy chain CDR2 region 97 2F7 Amino acid sequence of a FFYGSPYDFDV heavy chain CDR3 region 98 2F7 Amino acid sequence of a KASENVDTHTVS light chain CDR1 region 99 2F7 Amino acid sequence of a GTSNRFT light chain CDR2 region 100 2F7 Amino acid sequence of a GQSYRYPYT light chain CDR3 region 101 13A2 Amino acid sequence of a EYVMH heavy chain CDR1 region 102 13A2 Amino acid sequence of a GINPNTGGTTYNQKFEG heavy chain CDR2 region 103 13A2 Amino acid sequence of a RMPFGISAMDY heavy chain CDR3 region 104 13A2 Amino acid sequence of a SASSSVSYIH light chain CDR1 region 105 13A2 Amino acid sequence of a ETSKLAS light chain CDR2 region 106 13A2 Amino acid sequence of a HQWISYPPT light chain CDR3 region 107 14B3 Amino acid sequence of a DHAIH heavy chain CDR1 region 108 14B3 Amino acid sequence of a VINTYSGNTNFNQKLKG heavy chain CDR2 region 109 14B3 Amino acid sequence of a GAHSSGYPYWFYFDV heavy chain CDR3 region 110 14B3 Amino acid sequence of a RASSSVSYMHL light chain CDR1 region 111 14B3 Amino acid sequence of a ATSNLAS light chain CDR2 region 112 14B3 Amino acid sequence of a QQWSTNPFT light chain CDR3 region

Example 5

ADCC Activity Testing of Hybridoma Monoclonal Antibodies

[0071] ADCC (antibody-dependent cell-mediated cytotoxicity) activity of the antibodies was tested using the reporter gene method: 293T-B7H3, PC9 or DLD1 tumor cells were used as target cells, and Jurkat-mCD16.2-NF- κ B or Jurkat-hCD16-NF- κ B cells were used as effector cells. The activation signal of transcription factor NF- κ B mediated by mCD16.2 or hCD16 was used to screen ADCC-positive clones. 50 μ L of hybridoma antibodies that were positive in the above ELISA and FACS tests were mixed with 25 μ L of 293T-B7H3, PC9 or DLD1 tumor cells (7.5×10^4 cells/well), a resulting mixture was added to a 96-well plate, and then 25 μ L of Jurkat-mCD16.2-NF- κ B cells (2.5×10^4 cells/well) were added, mixed well, and incubated in an incubator for 4 hours at 37° C. 25 μ L of preheated Bright-Glo solution (Promega, Cat. No.: E2620) was added to each well and left standing for 3 minutes at room temperature in the dark, and the luminescence (Luminescence) signal value of each sample was measured by using a microplate reader (Tecan Spark).

[0072] Among hybridoma antibodies, mouse IgG1 antibodies had no ADCC activity. After gene sequencing, the inventors confirmed that among the 14 hybridoma antibodies, 9 hybridoma

antibodies were based on mouse IgG1, and 5 hybridoma antibodies were based on IgG2b or IgG2a (see Table 3). FIG. 6a shows ADCC reporter gene test results of hybridoma antibodies by using Jurkat-B7H3 as target cells. As expected, 5 hybridoma antibodies based on IgG2b or IgG2a had ADCC activity, while 9 hybridoma antibodies based on mouse IgG1 were inactive. Therefore, as shown in FIG. 6b, for 9 hybridoma antibodies based on mouse IgG1, the inventors conducted Fc replacement and turned these antibodies into chimeric antibodies based on human IgG1. FIG. 7 and FIG. 8 reflect the ADCC reporter gene test results by using PC9 or DLD1 tumor cells as targeted cells. The antibody was in the form of a chimeric antibody based on human IgG1. Based on the test results of ADCC reporter genes, antibodies with relatively strong ADCC activity included 9C8, 5B6, 7C9, F5, 7E6, 2A9, and 4F11.

TABLE-US-00006 TABLE 3 Fc Types of B7H3 Hybridoma Antibodies

Abs	2A9	13A2	4F11	9C8	5B6	7E6	7F5	2F7	7C9	2E10	7B7	14B3	2F12	15A2	Fc
	IgG2b	IgG2b	IgG1	IgG1	IgG1	IgG1	IgG1	IgG1	IgG1	IgG1	IgG1	IgG1	IgG1	IgG1	IgG1

Example 6

Humanizing B7H3 Antibodies 2A9, 4F11, 9C8, 15A2, 5B6 and 7B7

[0073] Considering the various characteristics of hybridoma antibodies, six B7H3 antibodies, namely 2A9, 4F11, 9C8, 15A2, 5B6, and 7B7, were selected for humanization.

[0074] Humanization of these B7H3 antibodies was achieved through the complementary-determining region grafting method. Firstly, IMGT database was queried for human germline antibody (germline antibody) sequences with the maximum sequence homology to the light chain and heavy chain variable regions of murine-derived antibodies 2A9, 4F11, 9C8, 15A2, 5B6 and 7B7. Germline IGKV3-11*01 was selected for humanization of the light chain variable region of the antibody 2A9, and IGHV1-2*02 was selected for humanization of the heavy chain variable region. Germline IGKV3-11*01 was selected for humanization of the light chain variable region of the antibody 4F11, and IGHV1-2*02 was selected for humanization of the heavy chain variable region. Germline IGKV3-11*01 was selected for humanization of the light chain variable region of the antibody 9C8, and IGHV1-46*01 was selected for humanization of the heavy chain variable region. Germline IGKV3-11*01 was selected for humanization of the light chain variable region of the antibody 15A2, and IGHV1-8*01 was selected for humanization of the heavy chain variable region. Germline IGKV1-39*01 was selected for humanization of the light chain variable region of the antibody 5B6, and IGHV1-69*02 was selected for humanization of the heavy chain variable region. Germline IGKV1-39*01 was selected for humanization of the light chain variable region of the antibody 7B7, and IGHV3-21*01 was selected for humanization of the heavy chain variable region. The CDR region of the murine-derived antibodies was retained, and the framework (framework) sequence of the murine-derived antibodies was replaced with a framework sequence of the human germline antibodies. A structural model of the murine antibodies was created to compare amino acids at each site within the framework regions of both humanized and murine antibodies. If the spatial structure of the CDR region was not damaged or changed when a human amino acid sequence was used at a specific site in the framework, the human amino acid sequence was used at the specific site, otherwise, the corresponding murine sequence was used at the site (i.e., a reverse mutation to the murine sequence).

[0075] Based on structural simulations, Ala at the 24.sup.th position of a humanized heavy chain of the 2A9 antibody was reversely mutated to Thr, Met at the 48.sup.th position was reversely mutated to Ile, Val at the 67.sup.th position was reversely mutated to Ala, Met at the 69.sup.th position was reversely mutated to Leu, Arg at the 71.sup.th position was reversely mutated to Val, and Thr at the 73.sup.th position was reversely mutated to Lys. Leu at 46.sup.th position of a humanized light chain of 2A9 antibody was reversely mutated to Arg, Leu at the 47.sup.th position was reversely mutated to Trp, Ile at the 48.sup.th position was reversely mutated to Val, and Phe at the 71.sup.th position was reversely mutated to Tyr. Ala at the 24.sup.th position of a humanized heavy chain of the antibody 4F11 was reversely mutated to Thr, Met at the 48.sup.th position was reversely

mutated to Ile, Val at the 67.sup.th position was reversely mutated to Ala, Met at the 69.sup.th position was reversely mutated to Leu, Arg at the 71.sup.st position was reversely mutated to Val, and Thr at the 73rd position was reversely mutated to Lys. Leu at 46.sup.th position of a humanized light chain of 4F11 antibody was reversely mutated to Arg, Leu at the 47.sup.th position was reversely mutated to Trp, Ile at the 48.sup.th position was reversely mutated to Val, and Phe at the 71.sup.st position was reversely mutated to Tyr. Met at the 48.sup.th position of a humanized heavy chain of the antibody 9C8 was reversely mutated to Ile, Val at the 67.sup.th position was reversely mutated to Ala, Met at the 69.sup.th position was reversely mutated to Leu, Arg at the 71.sup.st position was reversely mutated to Val, and Thr at the 73.sup.rd position was reversely mutated to Lys. Ile at 2.sup.nd position of a humanized light chain of 9C8 antibody was reversely mutated to Thr, Leu at the 46.sup.th position was reversely mutated to Arg, Leu at the 47.sup.th position was reversely mutated to Trp, and Phe at the 71.sup.st position was reversely mutated to Tyr. Met at the 48.sup.th position of a humanized heavy chain of the antibody 15A2 was reversely mutated to Ile, Val at the 67.sup.th position was reversely mutated to Ala, Met at the 69.sup.th position was reversely mutated to Leu, Arg at the 71.sup.st position was reversely mutated to Ala, and Thr at the 73.sup.rd position was reversely mutated to Lys. Tyr at 36.sup.th position of a humanized light chain of 15A2 antibody was reversely mutated to Phe, Leu at the 47.sup.th position was reversely mutated to Trp, Tyr at the 49.sup.th position was reversely mutated to His, Ile at the 58.sup.th position was reversely mutated to Phe, and Phe at the 71.sup.st position was reversely mutated to Tyr. Gly at the 27.sup.th position of a humanized heavy chain of 5B6 antibody was reversely mutated to Tyr, Ser at the 30.sup.th position was reversely mutated to Ile, Met at the 48.sup.th position was reversely mutated to Ile, Val at the 67.sup.th position was reversely mutated to Ala, and Ile at the 69.sup.rd position was reversely mutated to Leu. Leu at 46.sup.th position of a humanized light chain of 5B6 antibody was reversely mutated to Arg, Leu at the 47.sup.th position was reversely mutated to Pro, Gly at the 66.sup.th position was reversely mutated to Ala, Phe at the 71.sup.st position was reversely mutated to Tyr, and Phe at the 98.sup.th position was reversely mutated to Ile. Ser at the 49.sup.th position of a humanized heavy chain of 7B7 antibody was reversely mutated to Ala. Ile at 2.sup.nd position of a humanized light chain of 7B7 antibody was reversely mutated to Ser, and Ile at the 48.sup.th position was reversely mutated to Val.

[0076] The amino acid sequence numbers of the heavy chain and light chain variable regions of the humanized antibody are SEQ ID NO: 113 and SEQ ID NO: 114, respectively. The amino acid sequence numbers of the heavy chain and light chain variable regions of the humanized antibody 4F11 are SEQ ID NO: 115 and SEQ ID NO: 116, respectively. The amino acid sequence numbers of the heavy chain and light chain variable regions of the humanized antibody 9C8 are SEQ ID NO: 117 and SEQ ID NO: 118, respectively. The amino acid sequence numbers of the heavy chain and light chain variable regions of the humanized antibody 15A2 are SEQ ID NO: 119 and SEQ ID NO: 120, respectively. The amino acid sequence numbers of the heavy chain and light chain variable regions of the humanized antibody 5B6 are SEQ ID NO: 121 and SEQ ID NO: 122, respectively. The amino acid sequence numbers of the heavy chain and light chain variable regions of the humanized antibody 7B7 are SEQ ID NO: 123 and SEQ ID NO: 124, respectively. The above humanized antibodies were constructed as the IgG1 subtype. The amino acid sequence information of the variable regions of the heavy and light chains of humanized antibodies is shown in Table 4.

[0077] Nucleic acid sequences encoding the light and heavy chains of humanized antibodies 2A9, 4F11, 9C8, 15A2, 5B6 and 7B7 were synthesized and inserted into expression vector pcDNA3.1. 200 mL of 293 cells (cell density: 1×10^6 /mL) were co-transfected with 0.1 mg of antibody light chain expression plasmids and 0.1 mg of antibody heavy chain expression plasmids, and cultured while shaken in a shake flask at 37° C. for 6 days, and the supernatant was collected by centrifugation. Protein A was used to purify humanized antibodies, and the purified humanized antibodies were tested for activity.

TABLE-US-00007	TABLE 4	Sequence	Listing of	Humanized	B7H3	Antibodies	SEQ
Description	ID	Humanized	Sequence	NO.	Antibodies	description	Amino acid sequence
hu2A9	Humanized	QVQLVQSGAEVKKPGASVKV	SCKTS	heavy			
		GYTFTEYIMHWVRQAPGQGLEWIGGI	chain	NPNSGGTTYNEKFKGRATLTVDKSIST			
		AYMELSRLRSDDTAVYYCARRMPYW	YFDVWGQGTMTVTVSS	114	hu2A9	Humanized	
		EIVLTQSPATLSLSPGERATLSCSASS	light	chain	SVSYIHWYQQKPGQAPRRWVYDTSK		
		LASGIPARFSGSGSGTDYTLTISSELEP	EDFAVYYCQQWVSNPLTFGQGTKLEI	K	115	hu4F11	
		Humanized	QVQLVQSGAEVKKPGASVKV	SCKTS	heavy		
		GYTFTEYIMHWVRQAPGQGLEWIGGI	chain	NPNTGGITYNQKFKDRATLTVDKSIST			
		AYMELSRLRSDDTAVYYCARAGGNFL	WYFDVWGQGTMTVTVSS	116	hu4F11	Humanized	
		EIVLTQSPATLSLSPGERATLSCSASS	light	chain	SVSYMHYQKPGQAPRRWLYDTS		
		KLASGIPARFSGSGSGTDYTLTISSELEP	PEDFAVYYCQQWNSNPLTFGQGTKLEI	EIK	117	hu9C8	
		Humanized	QVQLVQSGAEVKKPGASVKV	SCKAS	heavy		
		GYTFTEYTIHWVRQAPGQGLEWIGGI	chain	NPNSGGTTYNLKFKGRATLTVDKSTS			
		TVYMELSSLRSEDVAVYYCARKEGFF	TTDYWGQGTMTVTVSS	118	hu9C8	Humanized	
		ETVLTQSPATLSLSPGERATLSCSASS	light	chain	SLSYIHWYQQKPGQAPRRWIYDTSNL		
		ATGIPARFSGSGSGTDYTLTISSELEPE	DFAVYYCQQWHYNPPTFGGGTKVEIK		119	hu15A2	
		Humanized	QVQLVQSGAEVKKPGASVKV	SCKAS	heavy		
		GYTFTSYDINWVRQATGQGLEWIGWI	chain	DPGDGSTKYNEKFKGRATLTANKSIST			
		AYMELSSLRSEDVAVYYCVRRGFDYW	GQGTMTVTVSS	120	hu15A2	Humanized	
		EIVLTQSPATLSLSPGERATLSCRTSSS	light	chain	VSYLHWFQQKPGQAPRLWIHDTSNL		
		ASGFARFSGSGSGTDYTLTISSELEPE	DFAVYYCHQRSSYPWTFGGGKVEIK		121	hu5B6	
		Humanized	QVQLVQSGAEVKKPGSSVKV	SCKAS	heavy		
		GYTFITYTIHWVRQAPGQGLEWIGYIN	chain	PTSDYANFNQKFKDRATLTADKSTSTA			
		YMELSSLRSEDVAVYYCARWGTTVFQ	FYFDYWGQGTMTVTVSS	122	hu5B6	Humanized	
		EIVLTQSPATLSLSPGERATLSCRASQ	light	chain	SLSYINWYQQKPGKAPKPIYATSSL		
		QSGVPSRFSGSASGTDTYTLTISLQ	EDFATYYCQHWSSDPPTIGGGTKVEI	K	123	hu7B7	
		Humanized	EVQLVESGGGLVKGPGSLRLSCAASG	heavy			
		FTFSSYAVSWVRQAPGKGLEWVASIS	chain	GGGIYIYYPDSVKGRFTISRDNANKNSL			
		YLQMNSLRSEDVAVYYCARHGGAGY	FDYWGQGTMTVTVSS	124	hu7B7	Humanized	
		DSQMTQSPSSLSASVGDRVTITCRGS	light	chain	ESVHSYLAHYQQKPGKAPKLLVYNAL		
		TLQSGVPSRFSGSGSGTDFTLTISLQ	PEDFATYYCQHYGSPPTFGGGTK	VEIK			

Example 7

Binding of Humanized B7H3 Antibodies to B7H3

[0078] The binding activity of humanized B7H3 antibody samples to B7H3 protein was assessed using ELISA and FACS, referring to the method detailed in Example 2. The results of humanized antibodies hu2A9, hu4F11, hu5B6, hu9C8, hu15A2 and hu7B7 are shown in Table 5 and Table 6 and FIG. 9 and FIG. 10. Overall, the humanized antibodies according to the present disclosure retained their high antigen affinity from before humanization, particularly notable in hu15A2, hu7B7, and hu5B6.

TABLE-US-00008 TABLE 5 Determination of Binding of Humanized B7H3 Antibodies to B7H3-mFc Protein via ELISA Antibody EC.sub.50 hu2A9 hu4F11 hu5B6 hu9C8 hu15A2 hu7B7 (ng/mL) 17.49 40.55 12.73 19.90 3.84 6.35

TABLE-US-00009 TABLE 6 Determination of Binding of Humanized B7H3 Antibodies to Cell Surface B7H3 via FACS Antibody EC.sub.50 hu2A9 hu4F11 hu5B6 hu9C8 hu15A2 Hu7B7 (ng/mL) 140.4 126.7 62.05 79.0 14.3 11.34

Example 8

Detection of Cellular Internalization of Humanized B7H3 Antibodies

[0079] For a specific measurement method, see Example 3. The results are shown in Table 7 and FIG. 11, showing that the humanized B7H3 antibodies had significant cellular internalization by

PC9 cells. Compared with the positive control groups (MGA018 from MacroGenics and 8H9 from MSKCC), the cellular internalization of the antibody in the present disclosure was much stronger than that of 8H9. In addition, hu15A2 and hu7B7 also outperformed MGC018.

TABLE-US-00010 TABLE 7 Cellular Internalization of Humanized B7H3 Antibodies by PC9 Tumor Cells

Antibody	EC.sub.50	hu2A9	hu4F11	Hu5B6	hu9C8	hu15A2	Hu7B7	MGC018	8H9
(ng/mL)	118.8	88.4	48.88	61.68	29.9.7	16.45	38.16	172.9	

Example 9

Cytotoxic Effect of Toxin-Conjugated Humanized B7H3 Antibodies (B7H3-ADC) on Calu-6 Tumor Cells

[0080] Vc-MMAE was conjugated to the B7H3 antibodies in a chemical conjugation method. TCEP was used to reduce B7H3 antibodies in a molar ratio of 3:1, and a resulting mixture was incubated at 37° C. for 45 minutes and subjected to ultrafiltration to remove TCE. The antibodies were resuspended in 0.5 mL of PBS, Vc-MMAE was added to the B7H3 antibodies in a mole ratio of 6 to 1, a resulting mixture was incubated at 4° C. for 120 minutes and subjected to ultrafiltration to remove unconjugated Vc-MMAE, and the antibody concentration was measured for later use. 100 μ L of PC9 (5000 cells per well) and 100 μ L serially diluted B7H3-ADC were added to a 96-well plate. After incubation at 37° C. for 5 days, x μ L of CellTiter-Glo was added to each well, fluorescence intensity was measured, and an inhibition ability of the antibody against tumor cell proliferation was calculated. The results are shown in FIG. 12 and Table 8. Humanized B7H3 antibodies conjugated with MMAE toxin had a significant toxic effect on Calu-6 tumor cells, and was more active than the positive control group (MGC018-MMAE) from MacroGenics.

TABLE-US-00011 TABLE 8 Cytotoxic Effect of Conjugate of Humanized B7H3 Antibodies and MMAE on Calu-6 Tumor Cells

Antibody	hu7B7-	hu15A2-	hu5B6-	hu2A9-	hu4F11-	hu9C8-	MGC018-	EC.sub.50	MMAE	MMAE	MMAE	MMAE	MMAE	MMAE	MMAE	MMAE	(pM)
	4.877	5.003	4.909	11.45	8.295	16.82	23.37										

Example 10

Detection of ADCC Effect of Humanized B7H3 Antibody

[0081] For a specific measurement method, see Example 5. The results are shown in Table 9 and FIG. 13, showing that the humanized B7H3 antibodies had significant ADCC activity. MGA271 (MG-Ab) is an antibody from MacroGenics. Compared with MGA271 (MG-Ab), humanized 9C8, 5B6, 2A9 and 4F11 antibodies had lower EC.sub.50 values, indicating better ADCC activity.

TABLE-US-00012 TABLE 9 ADCC Effect of Humanized B7H3 Antibody

Antibody	Internalization	EC.sub.50	hu2A9	hu4F11	Hu5B6	hu9C8	MGA271	Jurkat-B7H3	EC.sub.50		
	43.03	54.79	26.72	32.68	116	PC9 (ng/mL)	88.72	190.4	74.50	70.86	295.3

Example 11

ADCC Cytotoxic Effect of NK92MI-hCD16 Cells on Jurkat-B7H3 and PC9 Cells Mediated by Humanized B7H3 Antibodies

[0082] NK92MI-hCD16 was used to simulate NK cells to evaluate tumor-killing activity of B7H3 antibodies. 25 μ L of PC9 (0.5×10^5 cells per well) stained with CFSE and 25 μ L of NK92MI-hCD16 (1×10^5 cells per well) were mixed, and 50 μ L of serially diluted B7H3 antibodies was added. After the 96-well plate was incubated for 4 hours at 37° C., 5 μ L of 7-AAD was added to each well and the 96-well plate was incubated at room temperature for 10 minutes. The 96-well plate was centrifuged to remove the supernatant and resuspended. A flow cytometer was used to detect a ratio of cells positive for both 7-AAD and CFSE to CFSE-positive cells to obtain the B7H3 antibody-mediated NK cell killing activity. The results are shown in FIG. 14. Humanized B7H3 antibodies 9C8, 5B6, 2A9 and 4F11 could significantly cause an ADCC killing effect of NK92MI-hCD16 cells on PC9 tumor cells.

Example 12

Promoting Effect of Humanized B7H3 Antibodies on Secretion of Cytokines by Human PBMC Cells

[0083] For a specific measurement method, see Example 3. The results are shown in FIG. 15. Humanized B7H3 antibodies 5B6, 2A9 and 4F11 had a significant promoting effect on secretion of IFN- γ by human PBMC cells.

[0084] The foregoing embodiments are only preferred embodiments of the present disclosure and cannot be construed as a limitation on the protection scope of the present disclosure. That is, any simple equivalent changes and modifications made in accordance with the claims of the present disclosure and content of the utility model still fall within the protection scope of the present disclosure.

Claims

1. An anti-B7H3 monoclonal antibody or an antigen-binding fragment thereof, comprising a heavy chain CDR and a light chain CDR, wherein the heavy chain CDR and the light chain CDR are selected from one or more groups of CDR combinations denoted as a to n: TABLE-US-00013

Heavy chain CDR1	Heavy chain CDR2	Heavy chain CDR3	Light chain CDR1	Light chain CDR2	Light chain CDR3	Combination name
a	SEQ ID NO: 29	SEQ ID NO: 30	a	SEQ ID NO: 31	SEQ ID NO: 32	Heavy chain CDR1 CDR2 CDR3 a SEQ ID NO: 29 NO: 30 NO: 31 NO: 32 NO: 33 NO: 34
b	SEQ ID NO: 35	SEQ ID NO: 36	b	SEQ ID NO: 37	SEQ ID NO: 38	Heavy chain CDR1 CDR2 CDR3 b SEQ ID NO: 35 NO: 36 NO: 37 NO: 38 NO: 39 NO: 40
c	SEQ ID NO: 41	SEQ ID NO: 42	c	SEQ ID NO: 43	SEQ ID NO: 44	Heavy chain CDR1 CDR2 CDR3 c SEQ ID NO: 41 NO: 42 NO: 43 NO: 44 NO: 45 NO: 46
d	SEQ ID NO: 47	SEQ ID NO: 48	d	SEQ ID NO: 49	SEQ ID NO: 50	Heavy chain CDR1 CDR2 CDR3 d SEQ ID NO: 47 NO: 48 NO: 49 NO: 50 NO: 51 NO: 52
e	SEQ ID NO: 53	SEQ ID NO: 54	e	SEQ ID NO: 55	SEQ ID NO: 56	Heavy chain CDR1 CDR2 CDR3 e SEQ ID NO: 53 NO: 54 NO: 55 NO: 56 NO: 57 NO: 58
f	SEQ ID NO: 59	SEQ ID NO: 60	f	SEQ ID NO: 61	SEQ ID NO: 62	Heavy chain CDR1 CDR2 CDR3 f SEQ ID NO: 59 NO: 60 NO: 61 NO: 62 NO: 63 NO: 64
g	SEQ ID NO: 65	SEQ ID NO: 66	g	SEQ ID NO: 67	SEQ ID NO: 68	Heavy chain CDR1 CDR2 CDR3 g SEQ ID NO: 65 NO: 66 NO: 67 NO: 68 NO: 69 NO: 70
h	SEQ ID NO: 71	SEQ ID NO: 72	h	SEQ ID NO: 73	SEQ ID NO: 74	Heavy chain CDR1 CDR2 CDR3 h SEQ ID NO: 71 NO: 72 NO: 73 NO: 74 NO: 75 NO: 76
i	SEQ ID NO: 77	SEQ ID NO: 78	i	SEQ ID NO: 79	SEQ ID NO: 80	Heavy chain CDR1 CDR2 CDR3 i SEQ ID NO: 77 NO: 78 NO: 79 NO: 80 NO: 81 NO: 82
j	SEQ ID NO: 83	SEQ ID NO: 84	j	SEQ ID NO: 85	SEQ ID NO: 86	Heavy chain CDR1 CDR2 CDR3 j SEQ ID NO: 83 NO: 84 NO: 85 NO: 86 NO: 87 NO: 88
k	SEQ ID NO: 89	SEQ ID NO: 90	k	SEQ ID NO: 91	SEQ ID NO: 92	Heavy chain CDR1 CDR2 CDR3 k SEQ ID NO: 89 NO: 90 NO: 91 NO: 92 NO: 93 NO: 94
l	SEQ ID NO: 95	SEQ ID NO: 96	l	SEQ ID NO: 97	SEQ ID NO: 98	Heavy chain CDR1 CDR2 CDR3 l SEQ ID NO: 95 NO: 96 NO: 97 NO: 98 NO: 99 NO: 100
m	SEQ ID NO: 101	SEQ ID NO: 102	m	SEQ ID NO: 103	SEQ ID NO: 104	Heavy chain CDR1 CDR2 CDR3 m SEQ ID NO: 101 NO: 102 NO: 103 NO: 104 NO: 105 NO: 106
n	SEQ ID NO: 107	SEQ ID NO: 108	n	SEQ ID NO: 109	SEQ ID NO: 110	Heavy chain CDR1 CDR2 CDR3 n SEQ ID NO: 107 NO: 108 NO: 109 NO: 110 NO: 111 NO: 112

2. The anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof according to claim 1, further comprising a heavy chain variable region and a light chain variable region, wherein the heavy chain variable region and the light chain variable region are selected from one or more groups of variable region combinations denoted as A to N: TABLE-US-00014

Heavy chain variable region	Light chain variable region	Combination name
A	SEQ ID NO: 1	Heavy chain variable region A SEQ ID NO: 1
B	SEQ ID NO: 2	Heavy chain variable region B SEQ ID NO: 2
C	SEQ ID NO: 3	Heavy chain variable region C SEQ ID NO: 3
D	SEQ ID NO: 4	Heavy chain variable region D SEQ ID NO: 4
E	SEQ ID NO: 5	Heavy chain variable region E SEQ ID NO: 5
F	SEQ ID NO: 6	Heavy chain variable region F SEQ ID NO: 6
G	SEQ ID NO: 7	Heavy chain variable region G SEQ ID NO: 7
H	SEQ ID NO: 8	Heavy chain variable region H SEQ ID NO: 8
I	SEQ ID NO: 9	Heavy chain variable region I SEQ ID NO: 9
J	SEQ ID NO: 10	Heavy chain variable region J SEQ ID NO: 10
K	SEQ ID NO: 11	Heavy chain variable region K SEQ ID NO: 11
L	SEQ ID NO: 12	Heavy chain variable region L SEQ ID NO: 12
M	SEQ ID NO: 13	Heavy chain variable region M SEQ ID NO: 13
N	SEQ ID NO: 14	Heavy chain variable region N SEQ ID NO: 14

3. The anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof according to claim 2, wherein the heavy chain variable region and the light chain variable region are humanized.

4. The anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof according to claim 3, wherein the humanized heavy chain variable region and light chain variable region are selected from one or more groups of variable region combinations denoted as 1 to 6: TABLE-US-00015

Heavy chain variable region	Light chain variable region	Combination name
1	SEQ ID NO: 113	Heavy chain variable region 1 SEQ ID NO: 113
2	SEQ ID NO: 114	Heavy chain variable region 2 SEQ ID NO: 114
3	SEQ ID NO: 115	Heavy chain variable region 3 SEQ ID NO: 115
4	SEQ ID NO: 116	Heavy chain variable region 4 SEQ ID NO: 116
5	SEQ ID NO: 117	Heavy chain variable region 5 SEQ ID NO: 117
6	SEQ ID NO: 118	Heavy chain variable region 6 SEQ ID NO: 118

5. The anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof according to claim 4, further comprising a constant region of human IgG1.
 6. A method for using the anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof according to claim 1 for at least one of: regulating B7H3 activity or B7H3 level, selectively killing or inhibiting a tumor through toxin conjugation, mediating antibody-dependent cytotoxicity (ADCC), blocking an effect of B7H3 in immunosuppression to enhance body immunity, promoting T lymphocyte, or improving production of cytokine.
 7. A monoclonal antibody conjugate, comprising: a monoclonal antibody and a conjugating part, wherein the monoclonal antibody is the anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof according to claim 1, and wherein the conjugating part is selected from one or more at least one of a radionuclide, a medication, a toxin, a cytokine, a cytokine receptor fragment, an enzyme, fluorescein, or biotin.
 8. A method for using the monoclonal antibody conjugate according to claim 7 for at least one of: regulating B7H3 activity or B7H3 level, selectively killing or inhibiting a tumor through toxin conjugation, mediating antibody-dependent cytotoxicity (ADCC), blocking an effect of B7H3 in immunosuppression to enhance body immunity, promoting T lymphocyte, or improving production of cytokine.
 9. A method for using the monoclonal antibody conjugate according to claim 7 for preventing, treating, and/or adjuvantly treating a tumor.
 10. An expression vector, comprising a nucleic acid molecule of the anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof according to claim 1.
 11. A pharmaceutical composition, comprising: the anti-B7H3 monoclonal antibody or the antigen-binding fragment thereof according to claim 1; and a pharmaceutical carrier.
 12. The method of claim 6, wherein the cytokine is IFN- γ in T lymphocyte.
 13. The method of claim 8, wherein the cytokine is IFN- γ in T lymphocyte.
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